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CSE10

MODELLING AND SIMULATION

PRESENTED BY: JULIO CAESAR TADENA

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CSE10

COMPARATIVE ANALYSIS OF THE ULAS TO COASTAL ROADD FLYOVER ROUTE USING HYBRID SIMULATION

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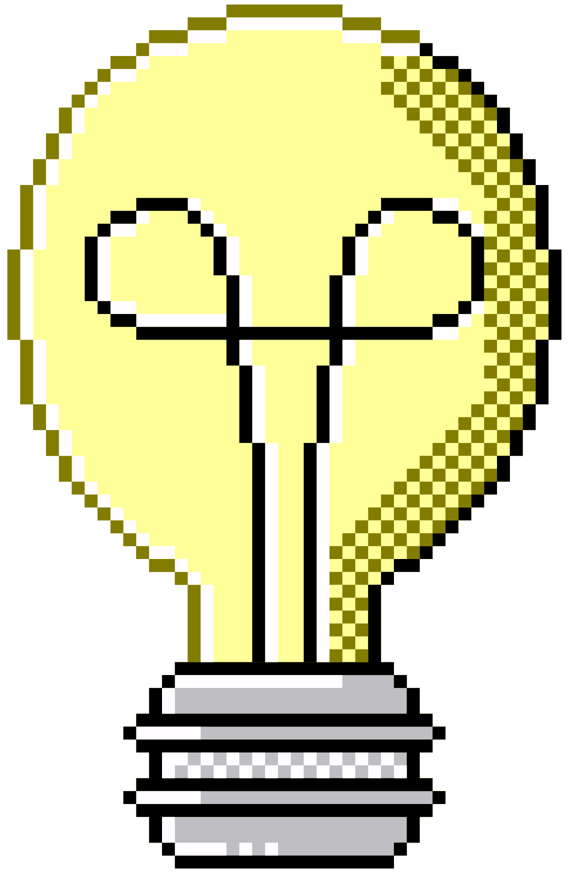
INTRODUCITON

- This project focuses on simulating the Ulas-to-Talomo Coastal Road Flyover system in Davao City using a hybrid discrete-event and spatial optimization approach to evaluate system performance and identify the most efficient route



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PROBLEM DEFINITION

- The Bottleneck: The Ulas-Talomo intersection in Davao City is a critical gateway heavily restricted by street-level traffic friction and multi-modal congestion.
- Proposed Infrastructure: Municipal planners have proposed an elevated, grade-separated flyover.
- Study Goal: Use OpenStreetMap (OSM) geographic data and stochastic load testing to mathematically capture delay patterns and optimize corridor throughput before physical construction.

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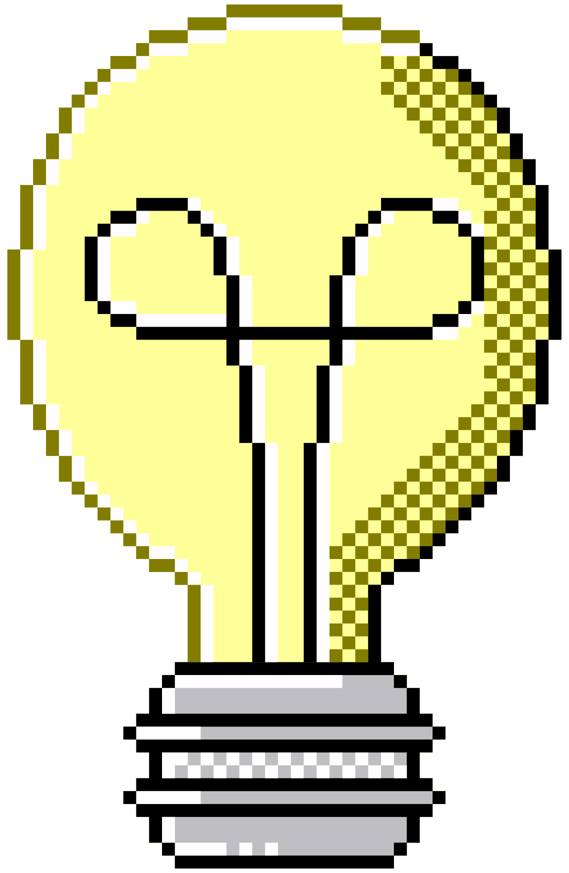
OBJECTIVES

- To model the Ulas-to-Talomo road network using a hybrid Deterministic-Stochastic simulation
- To measure average vehicle travel time and queue lengths at the flyover entry and exit points
- To analyze road and flyover capacity utilization
- To compare the performance of the government-proposed route against an algorithmically optimized route



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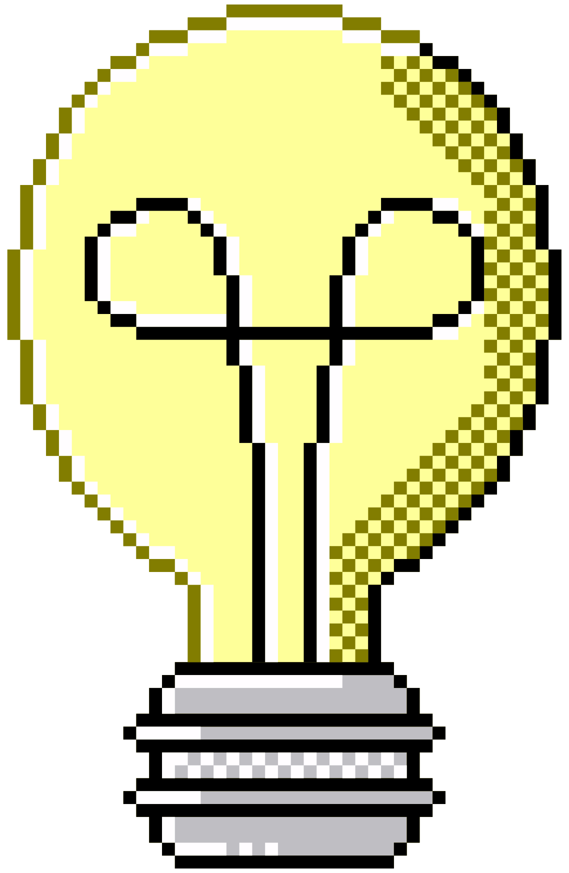
SYSTEM DESCRIPTION

- The system models a 2.3 km transit corridor in Davao City, stretching from the Ulas Junction (Source) to the Talomo Coastal Road Entry (Sink). The environment is represented as a Discrete Cost-Weighted Grid, where the geographic area is subdivided into a matrix of cells. Each cell serves as a decision point for the pathfinding algorithm, carrying specific "impedance" values based on its real-world composition

- Infrastructure (Roads): High-priority zones with the lowest traversal cost.
- Obstacles (Buildings): High-penalty zones to simulate the economic and social cost of demolition/displacement.
- Terrain (Natural Obstructions): Areas of high physical resistance (e.g., steep slopes or water bodies).

MODEL DESIGN

- The foundation of the model is a Discrete Cost-Surface Grid. Unlike standard vector maps, this model converts geographic reality into a computational matrix.



ASSUMPTIONS

- The following assumptions are integrated into the Python code to ensure the simulation remains valid and reproducible:

- Static Grid Resolution: The model assumes that a 100×100 resolution provides sufficient fidelity for a 2.3km stretch. Each cell is treated as a uniform unit of cost.
- Valid Geometry: We assume that OSM data with is_valid geometry tags correctly represent the physical boundaries of Davao’s buildings.
- Linear Cost Scaling: The model assumes that the cost of traversing through a building is exactly 20 times more "expensive" than traversing open land, representing the estimated ratio of construction vs. land acquisition/demolition.

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CONSTRAINTS

- The following constraints are integrated into the Python code to ensure the simulation remains valid and reproducible:

- **Boundary Limits**: The simulation is constrained to the bounding box of Latitudes 7.0430°N to 7.0590°N and Longitudes 125.5410°E to 125.5545°E.
- **No U-Turns**: The A* algorithm assumes a unidirectional flow from the source (Ulas) to the sink (Coastal Road); vehicles cannot change direction mid-path.
- **API Dependency**: The model design is constrained by the real-time availability and data density of the OpenStreetMap API.

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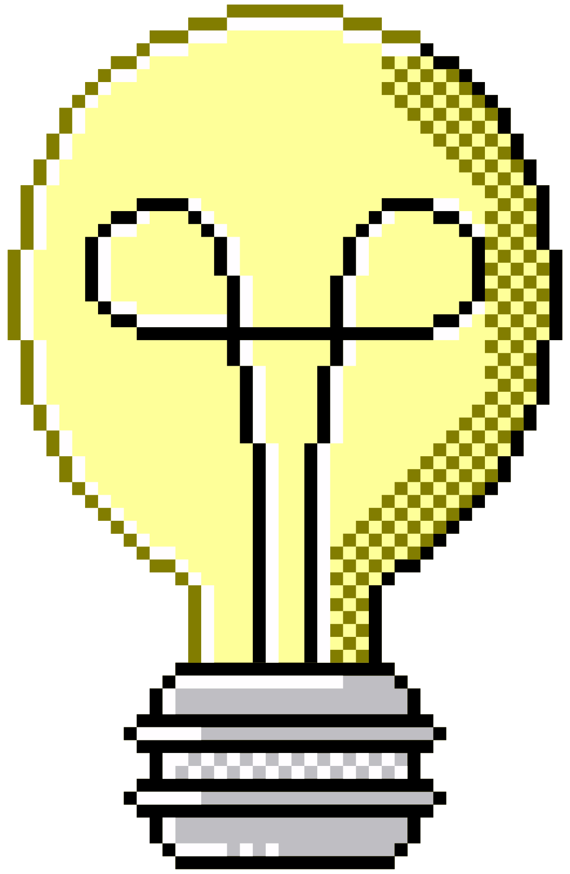
SIMULATION SETUP

- The simulation is constructed as a **Hybrid Spatial-Stochastic Model**. The environment is discretized into a rasterized grid to allow for non-linear path optimization.



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RESULTS AND ANALYSIS

Flyover	Distance	Avg Time	Max Delay
Real Flyover (Crimson)	1702.89m	104.74s	154.92s
Hypothetical Flyover (Green)	1310.45m	95.5s	140.86s
A* Ground Mesh (Blue)	2259.1m	290.47s	592.38s

- **Key Insights:** The real flyover requires 392 meters of additional pavement to follow the road right-of-way, yet it still captures 91% of the travel efficiency of a perfect straight line. The ground route completely choked, causing massive delay spikes and a deficit of several hundred uncompleted trips due to intersection gridlock.

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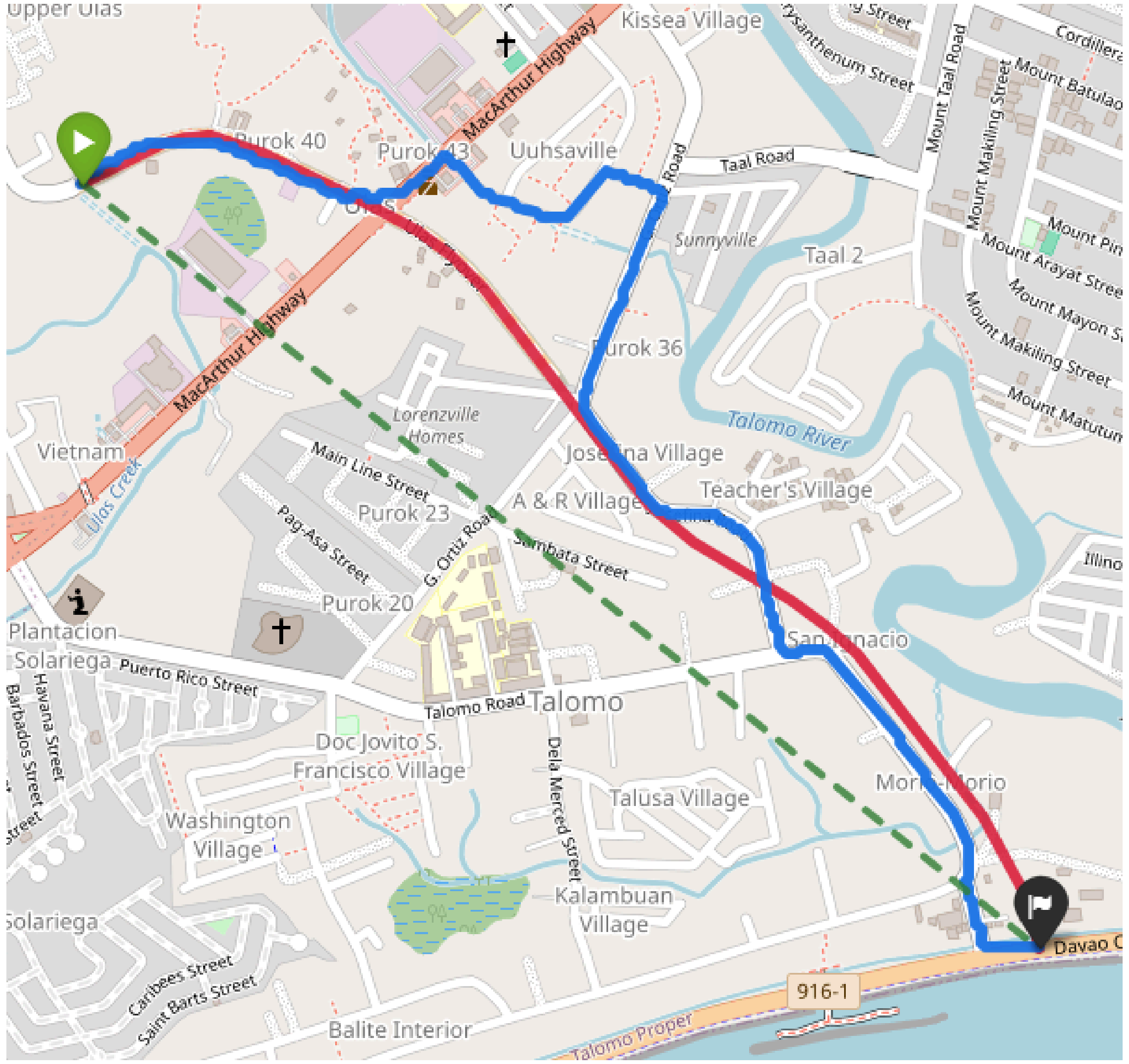
RESULTS AND ANALYSIS



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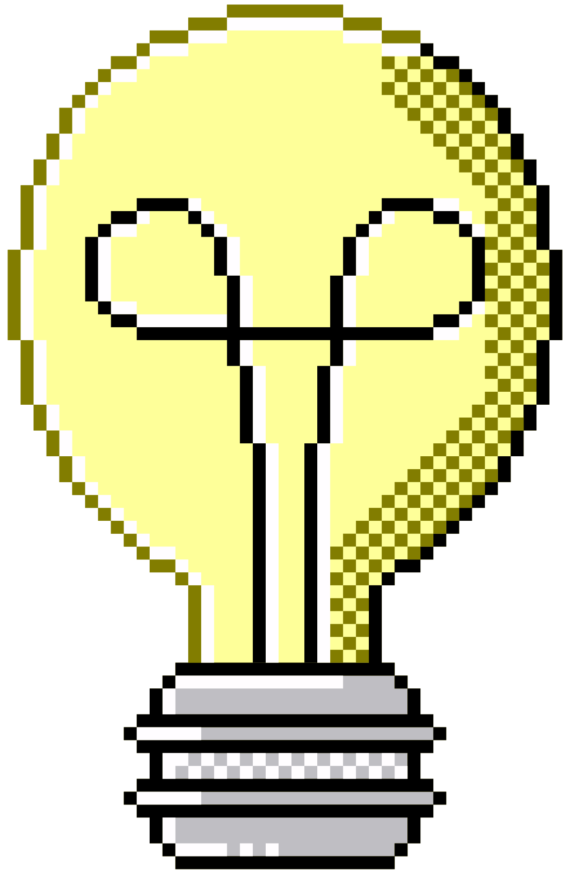
CONCLUSION

- **Infrastructure Justification**: Building the flyover saves an average of 3.1 minutes of delay per vehicle, translating to over 26.9 hours of collective human time saved per peak hour.



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RECOMMENDATIONS

- **Maximizing Design Efficiency:** To close the remaining 9.24-second gap to the ideal baseline, we recommend:

- **Roadway Super-Elevation:** Banking horizontal curves so vehicles maintain a constant 60 kph without tap-braking.
- **Active Lane Management:** Enforcing overtaking rules and peak-hour cargo restrictions to minimize micro-braking shockwaves.
- **Auxiliary Lanes:** Extending landing ramps further down to insulate the flyover from street-level tailbacks.
- **ITS Integration:** Deploying Variable Speed Signs and V2X infrastructure to transition traffic into a smooth, continuous stream.